Practice Quiz – **from Minimum Cost Flow Problem to Shortest Path**

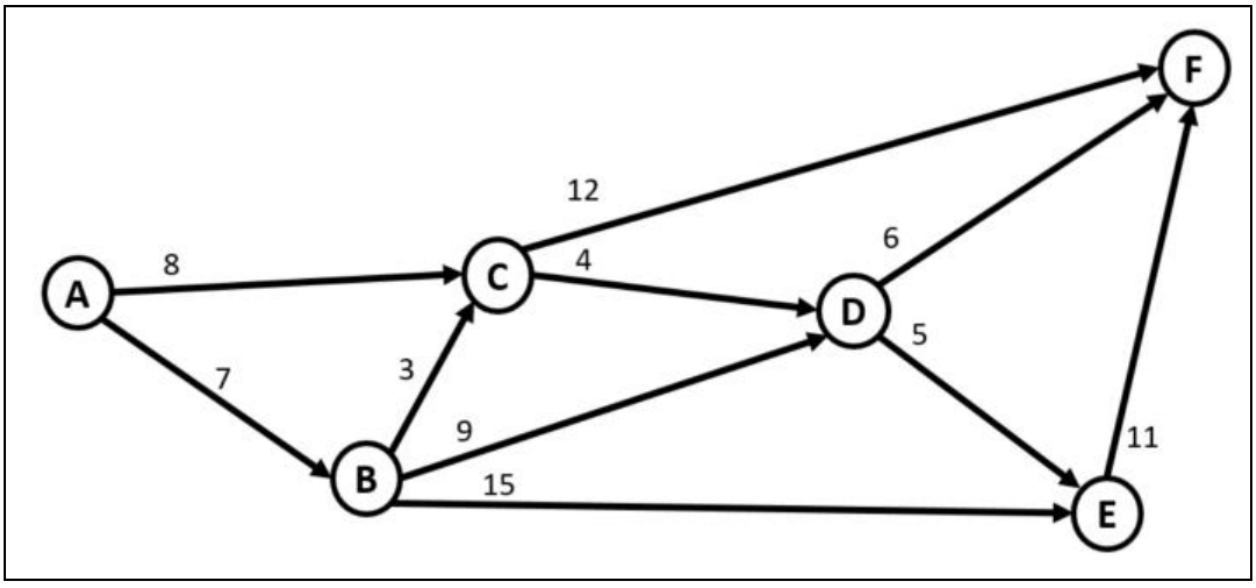

Figure 3a: Network Diagram

Addie wants to **find the shortest path from node A to node F** (Figure 3a) through linear programming (LP). Assume that decision variable Xij represents whether arc i to j is on the shortest path, for example, XAC represents whether arc A to C is on the shortest path.

1. State the total number of decision variables you need in the LP formulation of **the shortest path problem**. (1 mark)

   10 decision variables.

   XAB, XAC,

   XBC, XBD, XBE,

   XCD, XCF,

   XDE, XDF

   XEF

2. Construct the constraints for the following nodes based on the decision variables defined.
   i. Node A (2 marks)

      XAB+XAC = 1

   ii. Node C (2 marks)

      XCD + XCF – XAC – XBC = 0

   iii. Node F (2 marks)

      0 – XCF – XDF – XEF = -1

   iv. If the shortest path must pass node D, formulate this requirement as a constraint. (2 marks)

      XCD + XBD >=1 or

      XDF + XDE >=1